

International Marketing Analytics

MSc International Management / CEMS · WU Vienna · Winter 2026/27

AUTHOR

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 Important

Items marked **[to align]** are working assumptions. They must be aligned with the official course entry in the WU course catalogue (course 1243, winter term 2026) before publication.

Course at a glance

Course type	CEMS Hard Skills / Exclusive / Data Analytics, in person [to align: ECTS and contact hours]
Dates	Tuesdays, 6, 13, 20, 27 October and 3 November 2026 [to align: times and rooms]
Cohort	ca. 30 CEMS students
Language	English
Tools	Python 3.14 in Positron, Quarto, GitHub Desktop, DataCamp, AI assistants (Posit Assistant with GitHub Copilot)
Teaching data	Alpenglow, a fictional Vienna chocolate brand: six European markets in the labs, its web shop in Austria, Germany and Switzerland in the group case study
Assessment	DataCamp certificates 20 %, coding exercises 20 %, written exam 30 %, group case study report 30 %

Why this course

Marketing budgets are the largest discretionary spend in most consumer companies, and they are increasingly allocated across countries and media channels by models rather than by negotiation. Marketing mix modelling (MMM) has returned to the centre of this debate since privacy changes made user-level tracking and click-based attribution unreliable. Managers who can read, challenge and commission these models hold the budget conversation.

Marketing analytics is the measurement, management and analysis of marketing performance data to improve marketing effectiveness and return on investment. The course asks how price, place, product features and promotion drive sales, and how a fixed budget should be split across countries and channels. You will write little code by hand: AI assistants do most of the typing, and the course teaches you to specify, check and communicate analyses. Case examples include Zalando and NEOH.

Learning outcomes

On successful completion of this module, students will be able to:

Understand

1. Explain the fundamentals of marketing mix modelling, its strengths and limits.
2. Describe real-world applications of MMM in marketing.

Apply

3. Apply MMM in Python to cross-sectional and longitudinal data.
4. Interpret results and turn them into marketing decisions.

Transfer

5. Use AI tools for analysis and reporting, and check their output.
6. Think critically and work effectively in teams.

Prerequisites

None beyond bachelor-level statistics. No programming experience is expected. Bring your own laptop with administrator rights: we register the accounts and install the software together in Session 1, so no preparation is needed.

Schedule

Session	Date	Topic	Methods
1	Tue 6 Oct	Introduction and linear regression	Course content; registration and set-up (Python, Positron, GitHub Desktop, DataCamp); introduction to MMM; OLS, model fit and interpretation
2	Tue 13 Oct	Advanced regression	Log-log elasticities, dummy variables, non-

Session	Date	Topic	Methods
			linear effects, moderation and mediation, adstock and saturation, diagnostics
3	Tue 20 Oct	Logistic regression	Binary outcomes such as purchase or churn, odds ratios, model fit and classification
4	Tue 27 Oct	ARIMA	Trend, seasonality and autocorrelation; ARIMA and ARIMAX forecasts as the sales baseline
5	Tue 3 Nov	Advanced MMM tools	Bayesian MMM with PyMC-Marketing and Google Meridian; budget optimisation across channels and countries

Each session combines a concept lecture (60 min), a guided lab in Positron (90 min) and a team exercise (60 min). A short online quiz after each session shows you what to revise; it is not graded. Between sessions you read the assigned materials, work on the DataCamp courses and coding exercises, and write the case study report with your group. A guest speaker from industry joins one session **[to align]**.

Assessment

Component	Weight	Format
Online certificates	4 × 5 % = 20 %	individual: four assigned DataCamp Python courses, each certificate uploaded on Canvas by its deadline
Coding exercises	4 × 5 % = 20 %	individual: apply the code of a guided lab to a similar task; Python or Quarto file submitted on Canvas
Written exam	30 %	individual: 60 minutes, paper and pencil, multiple-choice questions on all five sessions, closed book [to align: date and room]

Component	Weight	Format
Group case study report	30 %	groups of at most 4: Quarto report on Alpenglow's web shop, submitted on Canvas by Monday 30 November 2026, midnight

Grades follow the WU scale: 1 Excellent (90 % and above), 2 Good (80 to 89 %), 3 Satisfactory (70 to 79 %), 4 Sufficient (60 to 69 %), 5 Fail (below 60 %). Attendance of at least 80 % of sessions is required **[to align]**.

Coding exercises

Each exercise takes the code from a guided lab and applies it to a similar task. The file must run from top to bottom. AI assistants are allowed; add a short prompt log at the end. This is individual work: discuss ideas, but write and check your own code. Deadlines are on Canvas.

Written exam

The exam checks understanding, not coding: concepts, interpretation and reading model output. The non-graded online quizzes use the same question format. Prepare with the session materials, the readings and the Q&A forum on Canvas.

Group case study report

The case study applies Sessions 1 to 4 to new data: Alpenglow's own web shop in Austria, Germany and Switzerland and its subscription club, the Alpenglow Club. Your group answers four questions from the Head of E-commerce:

Part	Question	Session	Points
A	What drives weekly orders? Data check, first regression, model fit, assumptions	1	15
B	How price-sensitive are customers in each country, and how does advertising work? Elasticities, temperature, adstock and saturation, moderation, mediation	2	30
C	Which subscribers are about to cancel, and who should get a	3	20

Part	Question	Session	Points
	retention offer? Logistic regression, classification, offer threshold		
D	How many orders should Germany plan for in Q1 2026? ARIMA versus ARIMAX on a holdout, forecast with intervals	4	20
E	Recommendations, limitations, reproducibility and presentation	all	15
Total			100

Each of the 20 criteria is scored Excellent (90 to 100 % of its points), Good (75 to 89 %), Adequate (60 to 74 %) or Insufficient (below 60 %). For every regression you check a few basic assumptions with the plots and numbers statsmodels already gives you; no advanced tests are required.

What you submit. One upload per group on Canvas, by **Monday 30 November 2026, midnight (23:59 Vienna time)**; the Canvas time stamp counts, and you can upload again until the deadline:

- `report.qmd` : one Quarto file with all code and text, which renders without errors from a fresh copy of your project folder;
- `report.html` : the rendered, self-contained report (no zip, no data).

Report rules. At most 3,000 words of text, excluding code, tables, figure captions and the appendix. All data work with polars, figures with plotnine, tables with great_tables and models with statsmodels. Every number in the text is computed in the report (inline code). The appendix holds the AI prompt log and a contribution statement signed by all members. If `report.qmd` does not render, the reproducibility criterion scores 0; if a number in the text does not match the code, its criterion is capped at *Adequate*.

Working together. Clone the course repository with GitHub Desktop and pull it for updates; copy the data and the template into your group's own project folder. How you combine your work is up to you: one member can keep the master file, or the group shares a private GitHub repository. Only the files uploaded to Canvas are assessed, not your commit history.

The brief with all tasks and the full assessment criteria, the data and the report template are in `case-study/` and `data/case/` of the course repository. Questions about the brief go to the Q&A forum on Canvas, so that every group gets the same answer.

Use of AI tools

AI assistants (Posit Assistant with GitHub Copilot, or similar) are expected and encouraged for all coding work.

Rules:

1. You are responsible for every number you report. Verify outputs with the checks taught in class.
2. Document the prompts that produced non-trivial code (a short prompt log in coding exercises; the appendix of the case study report).
3. The written exam is closed book and tests understanding, not code.
4. Fabricated references or results are academic misconduct under WU rules.

Reading

Session 1 reading (on Canvas):

- Fantini, F. and Narayandas, D. (2023). Analytics for marketers: When to rely on algorithms and when to trust your gut. *Harvard Business Review*, May–June 2023.

Core:

- Yildirim, G. and Kübler, R. V. (2025). *Applied Marketing Analytics Using Python*. SAGE. Chapters on regression, market response and marketing mix modelling.
- Schwarz, J. S., Chapman, C. and Feit, E. M. (2020). *Python for Marketing Research and Analytics*. Springer. Chapters 1 to 7 as Python and regression reference.

Marketing mix modelling:

- Chan, D. and Perry, M. (2017). Challenges and opportunities in media mix modeling. Google Research.
- Jin, Y., Wang, Y., Sun, Y., Chan, D. and Koehler, J. (2017). Bayesian methods for media mix modeling with carryover and shape effects. Google Research.
- Hanssens, D. M., Parsons, L. J. and Schultz, R. L. (2001). *Market Response Models: Econometric and Time Series Analysis*. Kluwer. Chapters 1 to 3 as background.
- Sethuraman, R., Tellis, G. J. and Briesch, R. A. (2011). How well does advertising work? Generalizations from meta-analysis of brand advertising elasticities. *Journal of Marketing Research*, 48(3), 457–471.
- PyMC Labs. *PyMC-Marketing documentation*; Google. *Meridian documentation*.

Attribution and experiments:

- Gordon, B. R., Zettelmeyer, F., Bhargava, N. and Chapsky, D. (2019). A comparison of approaches to advertising measurement: Evidence from big field experiments at Facebook. *Marketing Science*, 38(2), 193–225.
- Blake, T., Nosko, C. and Tadelis, S. (2015). Consumer heterogeneity and paid search effectiveness: A large-scale field experiment. *Econometrica*, 83(1), 155–174.
- Li, H. and Kannan, P. K. (2014). Attributing conversions in a multichannel online marketing environment: An empirical model and a field experiment. *Journal of Marketing Research*, 51(1), 40–56.

Course policies

- Bring your laptop and charger to every session; Session 1 includes the set-up.
- Coding exercises and the case study report are submitted on Canvas.
- Late submissions lose 10 percentage points per day unless agreed in advance **[to align]**.
- WU rules on academic integrity apply.

Contact

Dr Arne Floh, Institute for International Business, WU Vienna. Office hours: book online (link on Canvas). Questions about the course and the case study go to the Q&A forum on Canvas.